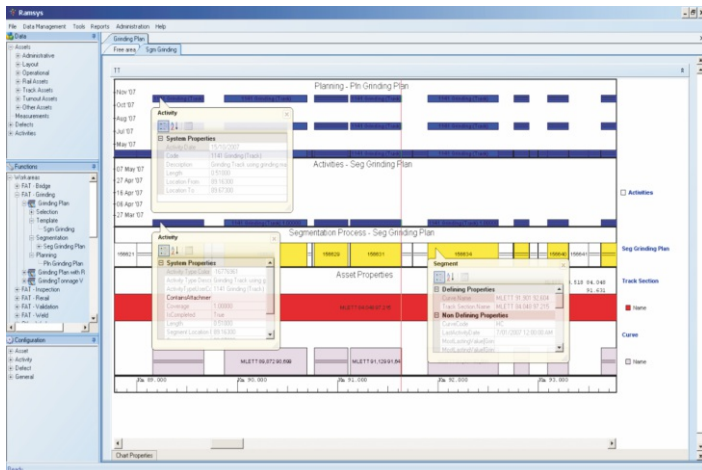


Decision Support System for Asset Management Planning



Working smarter, not harder - The increase of the railway systems availability, the improvement of safety features and the reduction of the maintenance costs strictly depend on the appropriate use of diagnostic methodologies and decision support systems for Infrastructure & Rolling Stock asset life-cycle management. RAMSYS represents the first and only Decision Support System for Railway Maintenance & Renewal Management. RAMSYS fully supports Condition-Based Maintenance and enables planning of maintenance interventions even before any faults come to light. Cyclic works and corrective maintenance can be significantly reduced because maintenance personnel can see today the future of their networks and fix problems before they happen. RAMSYS helps any railway to ensure the right work, at the right place, at the right time and for the right reason.





Maintenance and Renewal plans automatic generation

High quality levels of Infrastructures & Rolling Stock while reducing their cost of maintenance.

The railway consists of a great number of objects that are different in age, type and manufacture. These objects are submitted to different volume and degree of detrimental influences and react differently to a variety of remedial activities performed with different qualities.

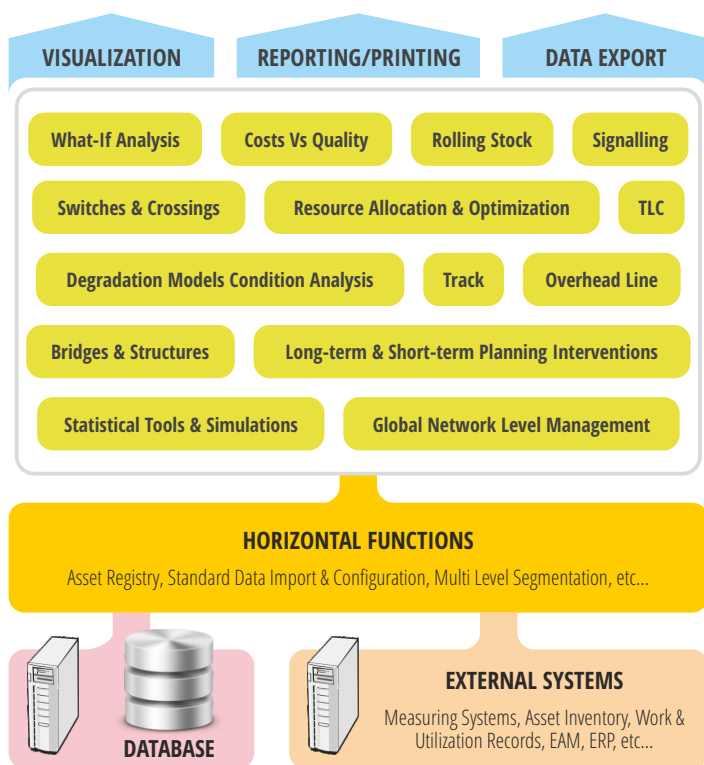
Management of maintenance and renewal works is therefore a huge and complex task involving several highly interrelated aspects such as infrastructure & rolling stock monitoring, railroad engineering and information systems. This necessitates a proper process and systems to cover comprehensive data management and analysis for decision support.

RAMSYS is a unique, integrated software platform for the management of all the data related to railway infrastructure & rolling stock condition based maintenance (e.g. assets, defects, measurements, activities, etc.) supporting M&R planning and control.

RAMSYS's ability to incorporate and process data of any kind, and any volume, into valuable information about railway assets conditions and behaviour helps shifting from corrective to condition based maintenance.

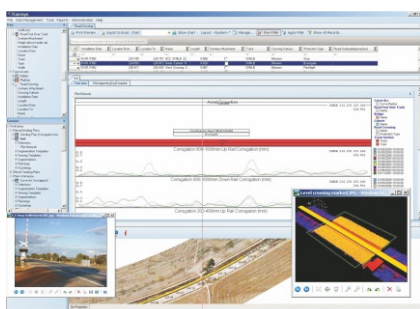
This makes RAMSYS a formidable weapon in the fight for increasing efficiency, availability, reliability and safety of infrastructures and trains, and helps railways keep control of maintenance budget.

Railway Asset Maintenance System Modular Architecture

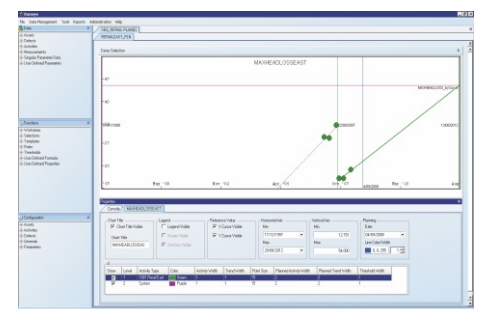
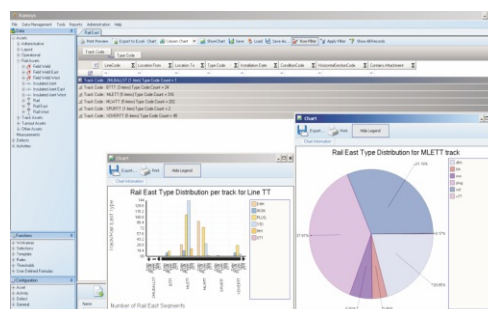


RAMSYS® features

- Ability to model railway infrastructure as linear/spatial assets using "link" and "node" objects; as well as "location-along-the-link" reference of all singular objects and "distributed" properties.
- Ability to handle any railway asset, such as rolling stock components (e.g. wheelsets, wheels, brakes, pantographs, etc.), switches & crossings, bridges and even the smallest items such as welds.
- Integrated handling capability of any kind of data different in nature (inventory, layout, condition, operation, work history, costs, etc.)
- Off line data collection from the field
- Superb visualization capabilities (e.g. GIS, track chart, data correlation in space and time, etc.)
- User tailored deterioration models & condition forecasting
- Automatic generation of maintenance and renewal plans based on a set of business rules
- M&R scenario simulation
- Full compatibility with external systems: EAM/ERP systems, GIS and other DBMS.



Integration of different visualisation modes



Standard reporting functionalities for railway maintenance decision support

Efficiently extracting useful information from data

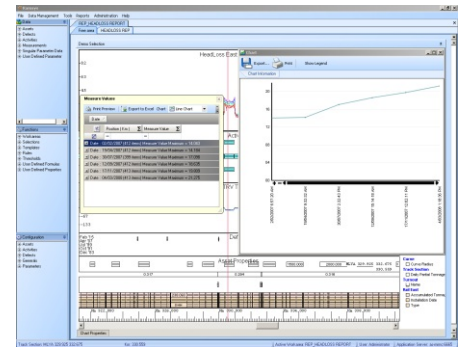
INPUT

- MEASUREMENTS
- ASSET INVENTORY
- WORK HISTORY
- INSPECTIONS
- COSTS



OUTPUT

- WORK ORDER PRIORITY LISTS
- CONDITION FORECASTING
- WHAT-IF ANALYSIS
- BUDGETING



Condition data trending

INPUT

Manage high Volumes of data from any source

With lots of data coming from numerous and often very different sources (on board and wayside diagnostic systems, EAM/ERP systems, etc.), it's easy to become overwhelmed and miss important aspects due to data that is rarely integrated. With its centralized data storage and analysis, RAMSYS is designed to provide your maintenance staff with all the tools needed to quickly and efficiently analyze terabyte of data.

OUTPUT

1 / Work Order Priority Lists

Thanks to its highly flexible structure RAMSYS can be adapted to the exact needs of your maintenance program, incorporating the specific experience already obtained on your infrastructure. The information allows RAMSYS to intelligently and automatically produce work order lists based on M&R priority, highlighting the need for maintenance activities exactly where they are needed, when they are needed. Your own M&R maintenance philosophies as well as governing body guidelines can be incorporated in RAMSYS.

2 / Condition Forecasting

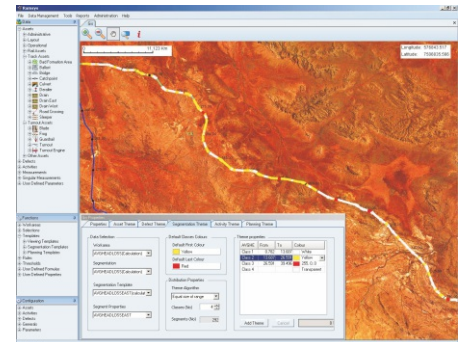
RAMSYS's ability to substitute all existing measuring/ monitoring software makes analysis, data correlation and defect finding and trending much easier for your maintenance engineers. On top of that RAMSYS is equipped with highly powerful deterioration models that are flexible and universal, helping you to forecast the future condition of your assets.

3 / Budgeting

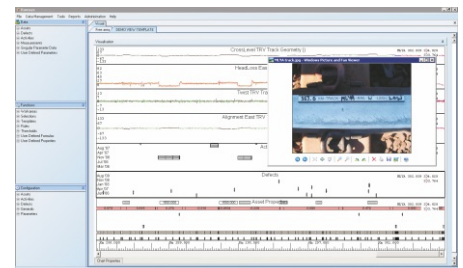
Taking into account work history, the cost of assets, level of defects, and even scenario simulation, RAMSYS can produce budgets based on actual and verifiable information, which can help you decide where to invest your resources.

4 / What-If Analysis

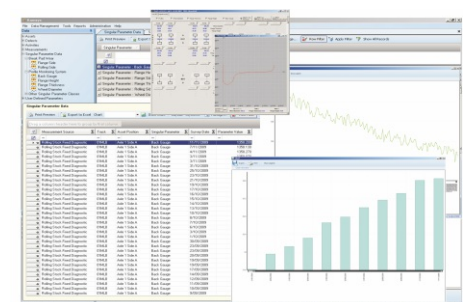
The what-if analysis engine allows your maintenance team to simulate various scenarios in the future, as the consequence of the asset's forecasted behavior and expected/simulated M&R strategy. This lets you test to find the perfect balance between M&R activities, their costs, and the resulting quality, availability and reliability of the assets.



Track-GPS(D) mapping and GIS thematic maps



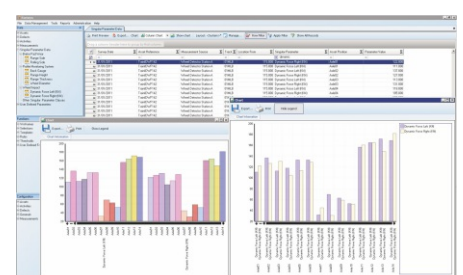
Linear visualisation for data correlation



Rolling Stock assets condition reporting and trending

Flexible Rule Engines, Intelligent Segmentation and Deterioration Models

RAMSYS's strengths are threefold: flexible rule creation allows your maintenance staff to model RAMSYS to your business requirements; intelligent data processing (e.g. dynamic data segmentation) helps you define your infrastructure into manageable segments; deterioration models help you understand and predict future condition of assets as well as to finally correlate rolling stock and infrastructure data. These ensure that all outputs are optimized for condition based maintenance.



Rolling Stock assets condition reporting and trending

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